

1. Loading the Palmer Penguins Dataset and Studying Its Structure (Dimensions, variable types, summary statistics)

code:

```
# Load the palmerpenguins package to access the dataset
library(palmerpenguins)

# Load the 'penguins' data frame into the environment
data("penguins")

# Display the first 6 rows of the dataset to get a quick look at the values
head(penguins)

# Provide a statistical summary of each variable (Min, Max, Mean, NAs, etc.)
summary(penguins)

# Display the internal structure of the object (data types and sample values)
str(penguins)

# Return the dimensions of the dataset (number of rows and number of columns)
dim(penguins)
```

Practical 1

```
> # Load the 'penguins' data frame into the environment
> data("penguins")
> # Display the first 6 rows of the dataset to get a quick look at the values
> head(penguins)
  species    island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
1  Adélie Torgersen   39.1         18.7           181             3750
2  Adélie Torgersen   39.5         17.4           186             3800
3  Adélie Torgersen   40.3         18.0           195             3250
4  Adélie Torgersen    NA           NA            NA              NA
5  Adélie Torgersen   36.7         19.3           193             3450
6  Adélie Torgersen   39.3         20.6           190             3650
  sex year
1  male 2007
2 female 2007
3 female 2007
4  <NA> 2007
5 female 2007
6  male 2007
```

```

> # Provide a statistical summary of each variable (Min, Max, Mean, NAs, etc.)
> summary(penguins)
  species      island  bill_length_mm bill_depth_mm flipper_length_mm
Adelie   :152  Biscoe   :168   Min.   :32.10   Min.   :13.10   Min.   :172.0
Chinstrap: 68  Dream    :124   1st Qu.:39.23   1st Qu.:15.60   1st Qu.:190.0
Gentoo   :124  Torgersen: 52   Median :44.45   Median :17.30   Median :197.0
                                     Mean   :43.92   Mean   :17.15   Mean   :200.9
                                     3rd Qu.:48.50   3rd Qu.:18.70   3rd Qu.:213.0
                                     Max.   :59.60   Max.   :21.50   Max.   :231.0
                                     NA's   :2       NA's   :2       NA's   :2

  body_mass_g      sex      year
Min.   :2700  female:165   Min.   :2007
1st Qu.:3550  male  :168   1st Qu.:2007
Median :4050  NA's  : 11   Median :2008
Mean   :4202                                     Mean :2008
3rd Qu.:4750                                     3rd Qu.:2009
Max.   :6300                                     Max.   :2009
NA's   :2
> # Display the internal structure of the object (data types and sample values)
> str(penguins)
'data.frame': 344 obs. of 8 variables:
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm  : num 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...

```

2. Detection of Missing Values in the Palmer Penguins

Dataset (Feature-wise missing value identification)

Code:

```
# Count missing values for each column
missing_counts <- colSums(is.na(penguins))
print(missing_counts)

# Total missing values in the entire dataset
sum(is.na(penguins))
```

Practical 2

```
> # Count missing values for each column
> missing_counts <- colSums(is.na(penguins))
> print(missing_counts)
      species      island  bill_length_mm  bill_depth_mm
      0         0         2              2
flipper_length_mm  body_mass_g      sex      year
      2         2         11         0
> # Total missing values in the entire dataset
> sum(is.na(penguins))
[1] 19
> |
```

3. Visualizing Feature Distribution through Histograms in the Palmer Penguins Dataset

(Feature-wise frequency distribution using Histograms)

Codes:

A: Features bill_length_mm

1. Bill Length: Check for normality or bimodal distribution

```
hist(penguins$bill_length_mm,  
col = "skyblue",  
main = "Distribution of Bill Length",  
xlab = "Bill Length (mm)",  
border = "white")
```

B: Features bill_depth_mm

2. Bill Depth: Check for skewness

```
hist(penguins$bill_depth_mm,  
col = "salmon",  
main = "Distribution of Bill Depth",  
xlab = "Bill Depth (mm)",  
border = "white")
```

C: Features flipper_length_mm

3. Flipper Length: Often shows distinct clusters in this dataset

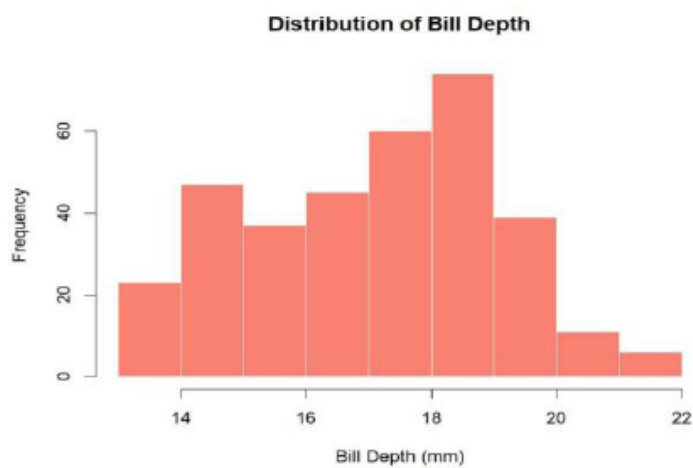
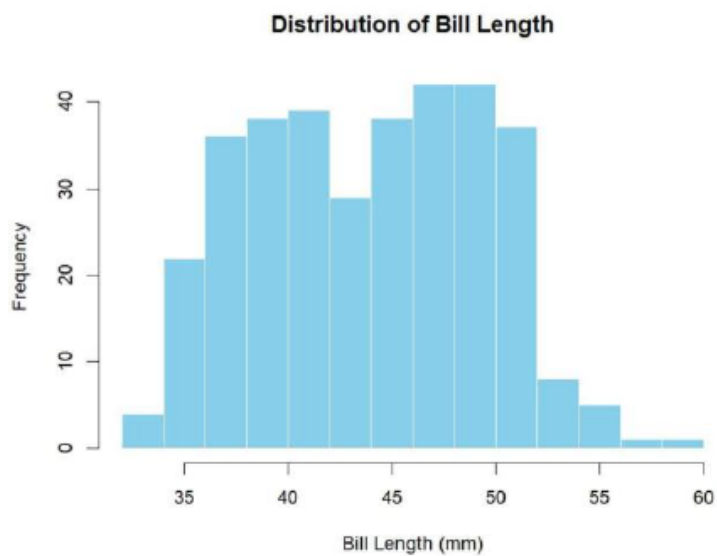
```
hist(penguins$flipper_length_mm,  
col = "seagreen",  
main = "Distribution of Flipper Length",  
xlab = "Flipper Length (mm)",  
border = "white")
```

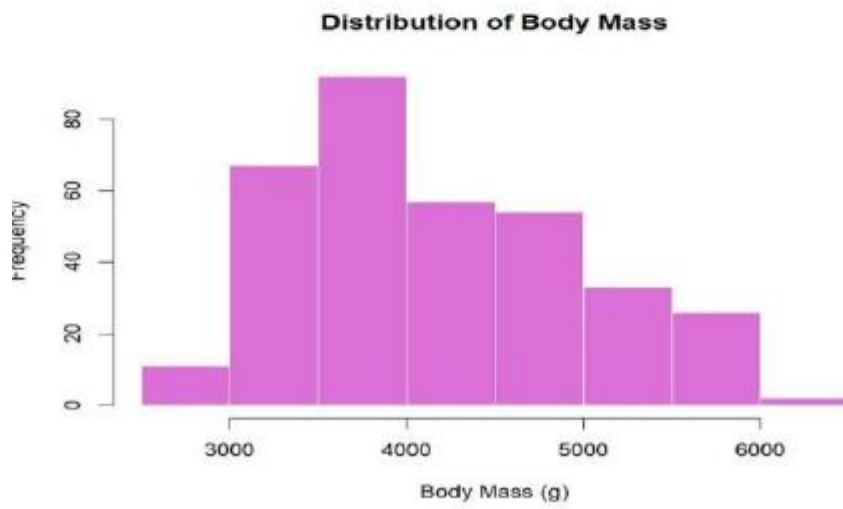
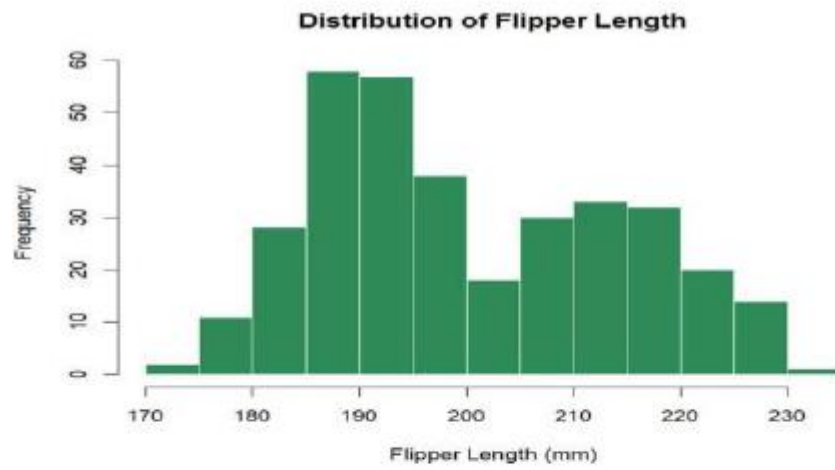
D: Features body_mass_g

4. Body Mass: Check for right-skewness (common in weight data)

```
hist(penguins$body_mass_g,  
col = "orchid",  
main = "Distribution of Body Mass",  
xlab = "Body Mass (g)",  
border = "white")
```

Practical 3





4. Handling Missing Values Using Mean and Median

Methods (Imputation and validation of results)

Code:

```
# --- Imputing Missing Values Using Median ---  
  
# We use the Median because these features show skewness and non-normal distributions.  
  
# The median is a robust measure that is less affected by outliers and skewness compared to  
the mean.  
  
# Impute bill_length_mm with median  
penguins$bill_length_mm[is.na(penguins$bill_length_mm)] <-  
median(penguins$bill_length_mm, na.rm = TRUE)  
  
# Impute bill_depth_mm with median  
penguins$bill_depth_mm[is.na(penguins$bill_depth_mm)] <-  
median(penguins$bill_depth_mm, na.rm = TRUE)  
  
# Impute flipper_length_mm with median  
penguins$flipper_length_mm[is.na(penguins$flipper_length_mm)] <-  
median(penguins$flipper_length_mm, na.rm = TRUE)  
  
# Impute body_mass_g with median  
penguins$body_mass_g[is.na(penguins$body_mass_g)] <- median(penguins$body_mass_g,  
na.rm = TRUE)  
  
# --- Validation: Check if NAs still exist ---  
  
# These should all return 0 if the imputation was successful.  
  
sum(is.na(penguins$bill_length_mm))  
sum(is.na(penguins$bill_depth_mm))  
sum(is.na(penguins$flipper_length_mm))  
sum(is.na(penguins$body_mass_g))
```

Practical 4

```
> # --- Imputing Missing Values Using Median ---
> # Median is used because the features are skewed and non-normally distributed.
> # It is robust to outliers compared to the mean.
>
> # Impute bill_length_mm with median
> penguins$bill_length_mm[is.na(penguins$bill_length_mm)] <-
+   median(penguins$bill_length_mm, na.rm = TRUE)
>
> # Impute bill_depth_mm with median
> penguins$bill_depth_mm[is.na(penguins$bill_depth_mm)] <-
+   median(penguins$bill_depth_mm, na.rm = TRUE)
>
> # Impute flipper_length_mm with median
> penguins$flipper_length_mm[is.na(penguins$flipper_length_mm)] <-
+   median(penguins$flipper_length_mm, na.rm = TRUE)
>
> # Impute body_mass_g with median
> penguins$body_mass_g[is.na(penguins$body_mass_g)] <-
+   median(penguins$body_mass_g, na.rm = TRUE)
>
> # --- Validation: Check if NAs still exist ---
> sum(is.na(penguins$bill_length_mm))
[1] 0
> sum(is.na(penguins$bill_depth_mm))
[1] 0
> sum(is.na(penguins$flipper_length_mm))
[1] 0
> sum(is.na(penguins$body_mass_g))
[1] 0
```

5. Performing Basic Statistical Analysis on Palmer Penguins

Data (Mean, Median, Mode, Range, Min, Max, Standard Deviation)

Code:

```
# Statistics for the 'body_mass_g' (Body Mass of Species)
mean_val <- mean(penguins$body_mass_g)
median_val <-
median(penguins$body_mass_g) sd_val <-
sd(penguins$body_mass_g) range_val <-
range(penguins$body_mass_g)
min_max <- c(min(penguins$body_mass_g), max(penguins$body_mass_g))
# Displaying results
cat("Mean:", mean_val, "\nMedian:", median_val, "\nStd Dev:", sd_val)
```

Practical 5

```
> mean_val <- mean(penguins$body_mass_g)
> median_val <- median(penguins$body_mass_g)
> sd_val <- sd(penguins$body_mass_g)
> range_val <- range(penguins$body_mass_g)
> min_max <- c(min(penguins$body_mass_g), max(penguins$body_mass_g))
> # Displaying results
> cat("Mean:", mean_val, "\nMedian:", median_val, "\nStd Dev:", sd_val)
Mean: 4200.872
Median: 4050
Std Dev: 799.6965
```

6. Box Plot Visualization of Palmer Penguins Dataset Features with Interpretation

(Spread of data and outlier identification)

Codes:

A: Features bill_length_mm

--- Visualizing Distribution with Boxplot ---

Since the data is skewed and non-normal, a boxplot is an excellent choice.

It clearly shows the Median (the thick line in the middle), which we used for imputation.

```
boxplot(penguins$bill_length_mm,
```

```
main = "Boxplot of Bill Length of penguins",
```

```
ylab = "Bill Length of penguins (mm)",
```

```
col = "skyblue", # Filling with skyblue for clarity
```

```
border = "darkblue", # Dark border for better contrast
```

```
# 'notch = TRUE' helps visualize the confidence interval around the median.
```

```
# If notches of two plots don't overlap, it's strong evidence that medians differ.
```

```
notch = TRUE,
```

```
frame = TRUE, # Adds a frame around the plot
```

```
cex.axis = 1.2, # Increase font size for axis labels
```

```
cex.lab = 1.3, # Increase font size for axis titles
```

```
cex.main = 1.4) # Increase font size for the main title
```

B: Features bill_depth_mm

--- Visualizing Distribution with Boxplot ---

Since the data is skewed and non-normal, a boxplot is an excellent choice.

It clearly shows the Median (the thick line in the middle), which we used for imputation.

```
boxplot(penguins$bill_depth_mm,
```

```
main = "Boxplot of Bill Depth of penguins",
```

```
ylab = "Bill Depth of penguins (mm)",
```

```
col = "pink", # Filling with skyblue for clarity
```

```
border = "red", # Dark border for better contrast
# 'notch = TRUE' helps visualize the confidence interval around the median.
# If notches of two plots don't overlap, it's strong evidence that medians differ.
notch = TRUE,
frame = TRUE, # Adds a frame around the plot
cex.axis = 1.2, # Increase font size for axis labels
cex.lab = 1.3, # Increase font size for axis titles
cex.main = 1.4) # Increase font size for the main title
```

C: Features flipper_length_mm

```
# --- Visualizing Distribution with Boxplot ---
# Since the data is skewed and non-normal, a boxplot is an excellent choice.
# It clearly shows the Median (the thick line in the middle), which we used for imputation.
boxplot(penguins$flipper_length_mm,
main = "Boxplot of Flipper Length of penguins",
ylab = "Flipper Length of penguins (mm)",
col = "purple", # Filling with skyblue for clarity
border = "black", # Dark border for better contrast
# 'notch = TRUE' helps visualize the confidence interval around the median.
# If notches of two plots don't overlap, it's strong evidence that medians differ.
notch = TRUE,
frame = TRUE, # Adds a frame around the plot
cex.axis = 1.2, # Increase font size for axis labels
cex.lab = 1.3, # Increase font size for axis titles
cex.main = 1.4) # Increase font size for the main title
```

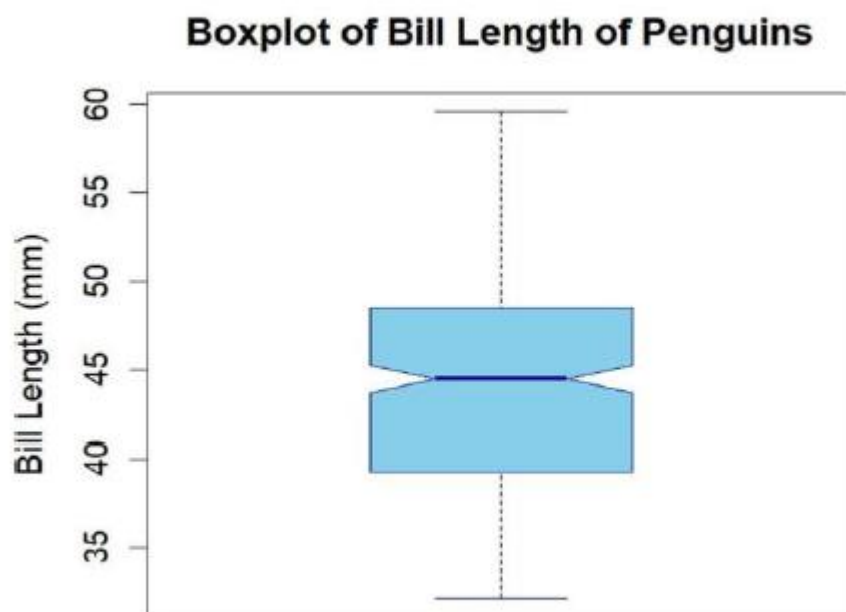
D: Features body_mass_g

```
# --- Visualizing Distribution with Boxplot ---
# Since the data is skewed and non-normal, a boxplot is an excellent choice.
```

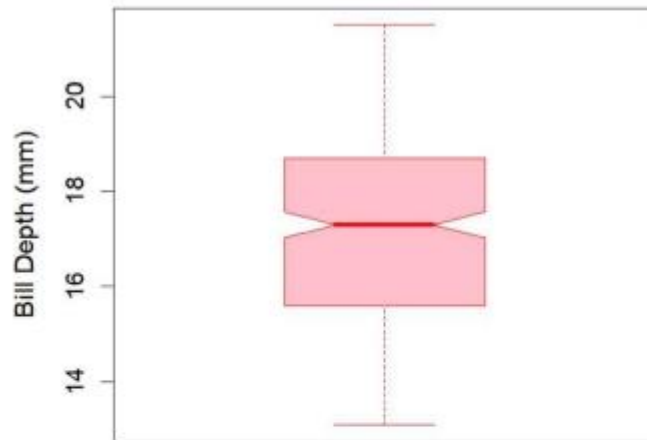
It clearly shows the Median (the thick line in the middle), which we used for imputation.

```
boxplot(penguins$body_mass_g,  
main = "Boxplot of Body Mass of penguins",  
ylab = "Body Mass of penguins (g)",  
col = "skyblue", # Filling with skyblue for clarity  
border = "darkblue", # Dark border for better contrast  
# 'notch = TRUE' helps visualize the confidence interval around the median.  
# If notches of two plots don't overlap, it's strong evidence that medians differ.  
notch = TRUE,  
frame = TRUE, # Adds a frame around the plot  
cex.axis = 1.2, # Increase font size for axis labels  
cex.lab = 1.3, # Increase font size for axis titles  
cex.main = 1.4) # Increase font size for the main title
```

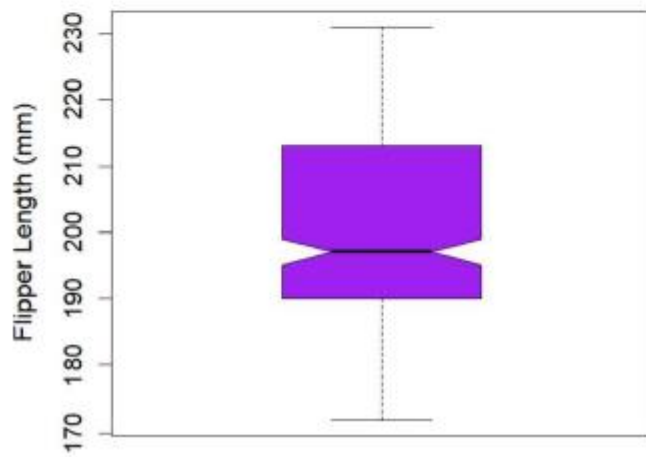
Practical 6



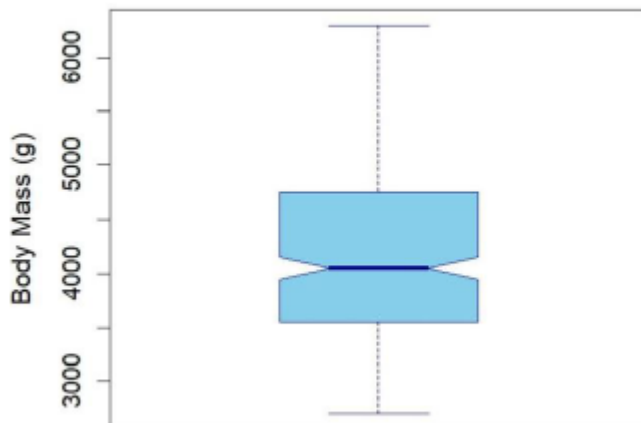
Boxplot of Bill Depth of Penguins



Boxplot of Flipper Length of Penguins



Boxplot of Body Mass of Penguins



7. Scatter Plot Analysis Between Selected Palmer Penguins

DataSet Variables

(Relationship analysis with proper interpretation)

Codes:

A: Features bill_length_mm vs bill_depth_mm

Visualizing the relationship between Bill Length and Bill Depth

```
plot(penguins$bill_length_mm, penguins$bill_depth_mm,
```

```
xlab = "Bill Length of penguins (mm)", # Label for the x-axis
```

```
ylab = "Bill Depth of penguins (mm)", # Label for the y-axis
```

```
main = "Scatter Plot: bill_length_mm vs bill_depth_mm", # Plot title
```

```
pch = 19, # Solid circle symbol for points
```

```
col = "darkgreen") # Color of the points
```

Statistical Note:

Look for clusters! While the overall trend might look slightly negative

or messy, the trend within individual species is usually positive.

B: Features flipper_length_mm vs body_mass_g

Visualizing the relationship between Bill Length and Bill Depth

```
plot(penguins$flipper_length_mm, penguins$body_mass_g,
```

```
xlab = "Flipper Length of penguins (mm)", # Label for the x-axis
```

```
ylab = "Body Mass of penguins (g)", # Label for the y-axis
```

```
main = "Scatter Plot: flipper_length_mm vs body_mass_g", # Plot title
```

```
pch = 19, # Solid circle symbol for points
```

```
col = "salmon") # Color of the points
```

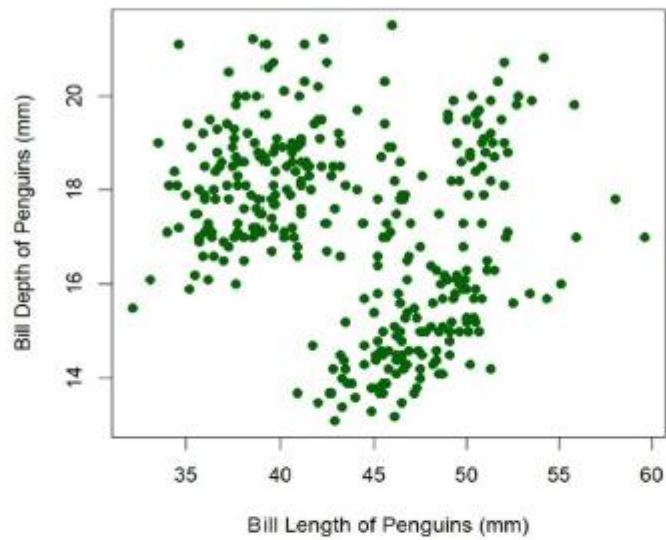
Statistical Note:

Look for clusters! While the overall trend might look slightly negative

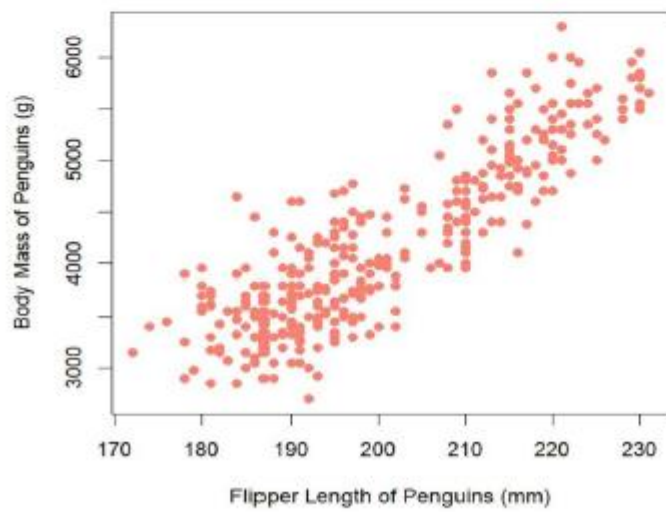
or messy, the trend within individual species is usually positive.

Practical 7

Scatter Plot: Bill Length vs Bill Depth



Scatter Plot: Flipper Length vs Body Mass



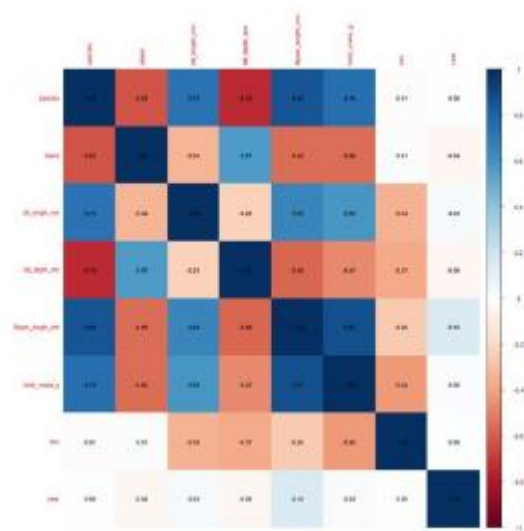
8. Heatmap Visualization for Correlation Analysis of Palmer Penguins Features

(Understanding positive and negative correlations)

Code:

```
#change categories → numbers values
# for sex feature
penguins$sex <- ifelse(penguins$sex == "male", 0, 1)
#for species feature
penguins$species <- as.numeric(
  factor(penguins$species,
    levels = c("Adelie", "Chinstrap", "Gentoo")))
)
#for island feature
penguins$island <- as.numeric(
  factor(penguins$island,
    levels = c("Biscoe", "Dream", "Torgersen")))
)
#Remove sex features NA values
library(dplyr)
penguins <- penguins %>% filter(!is.na(sex))
# 2. Generate the Correlation Heatmap
library(corrplot)
cor_matrix <-
cor(penguins)
corrplot(cor_matrix,
  method = "color",
  type = "full",
  tl.cex = 0.8,
  number.cex = 0.7)
```

Practical 8



9. Bubble Chart Representation of MTCARS Data with Interpretation

(Multi-variable visualization using size and color)

Codes:

A.Features, weights, and miles per gallon

```
library(readxl)
```

```
library(ggplot2)
```

```
df<-read.csv("C:/Users/hasan/Downloads/mtcars.csv")
```

```
summary(df)
```

```
ggplot(mtcars, aes(
```

```
  x = wt, # Weight as the x-axis
```

```
  y = mpg, # MPG as the y-axis
```

```
  size = hp, # Horsepower determines bubble size
```

```
  label = hp # Label showing Horsepower value
```

```
)) +
```

```
  geom_point(alpha = 0.6, color = "red") +
```

```
  # Using check_overlap = TRUE to keep the plot clean
```

```
  geom_text(vjust = -0.7, size = 3, check_overlap = TRUE) +
```

```
  scale_size(range = c(5, 25)) +
```

```
  labs(
```

```
    title = "Bubble Plot of Mtcars Data",
```

```
    x = "Weight (1000 lbs)",
```

```
    y = "Miles Per Gallon",
```

```
    size = "Horsepower"
```

```
  ) +
```

```
  theme_minimal()
```

B.Features Engine Displacement and Quarter Mile

Time # Bubble plot with your chosen features

```
ggplot(mtcars, aes(x = disp, # X-axis: engine displacement
```

```
y = qsec, # Y-axis: quarter-mile time
size = mpg, # Bubble size: fuel efficiency
color = factor(am)) + # Bubble color: transmission (0=Auto, 1=Manual)
geom_point(alpha = 0.6) +
scale_size(range = c(3, 12)) + # Adjust bubble size range
labs(title = "Bubble Plot of mtcars",
x = "Engine Displacement (cu.in.)",
y = "Quarter Mile Time (sec)",
size = "Miles per Gallon",
color = "Transmission\n(0=Auto, 1=Manual)") +
theme_minimal()
```

Practical 9

